

1. Prototype Overview

1. Prototype Summary

A functional greenhouse lighting prototype (Fig. 1) was fabricated using a Raspberry Pi Pico RP2040, two buttons, and three LEDs (Red, Blue, Green). The circuit, assembled on a breadboard with standard resistors, is housed in a custom 3D-printed PLA enclosure secured by nuts and bolts (Fig. 2).

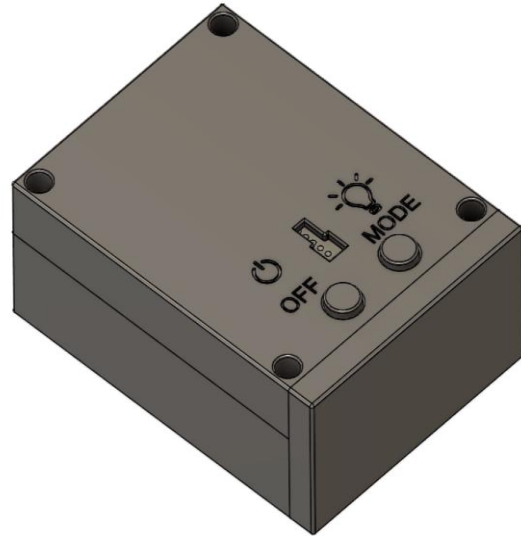
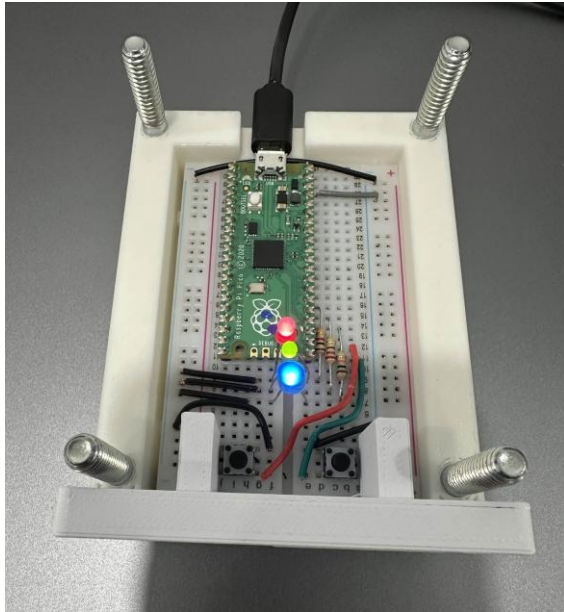


Fig. 1. Prototype of the greenhouse light is represented by the a) physical and b) CAD models.

The prototype logic (Fig. 2a) uses a finite state machine with three modes: OFF, PLANTS ONLY (Red/Blue), and PLANTS & PEOPLE (White). Users transition between states using the 'OFF' and 'MODE' buttons. This logic is processed by the microcontroller (*see 04.2_Software_SourceCode*) to control the physical circuit shown in Fig. 2b.

2. Design Process

The design process involved iterative exploration of the design space, moving from broad concept sketches to refined CAD modelling and code implementation.

- We initially explored a **magnetic clamshell design** to prioritize tool-less disassembly (Objective O-3.1-STRU-3), as seen in our concept generation sketches (*see 04.1_Concept_Brainstorming_Sketches.png*). However, due to stock shortages of neodymium magnets at MYFab, we pivoted to a hand-bolted design. While this required a redesign of the CAD chassis (*see 04.2_Structural_CAD_Environment.png*), it ultimately provided robust structural rigidity for the breadboard housing.
- For space above the breadboard for electrical components, we increased the depth of the box. This created the challenge to ensure buttons were accessible through the case. We addressed this with **3D-printed buttons** (*see 04.1_Mechanism_Design_ButtonPlungers.png*).
- The control logic evolved into a non-blocking state machine (*see 04.2_Software_SourceCode.py*). During electrical testing, we discovered that the range of sample data provided in the LED datasheet was insufficient for calculating the relevant parameters using

larger resistors. We debugged this by **manually testing resistors** while verifying the current and voltages remained within a safe range for the components (see 04.3_Debug_ResistorFix_Log.png).

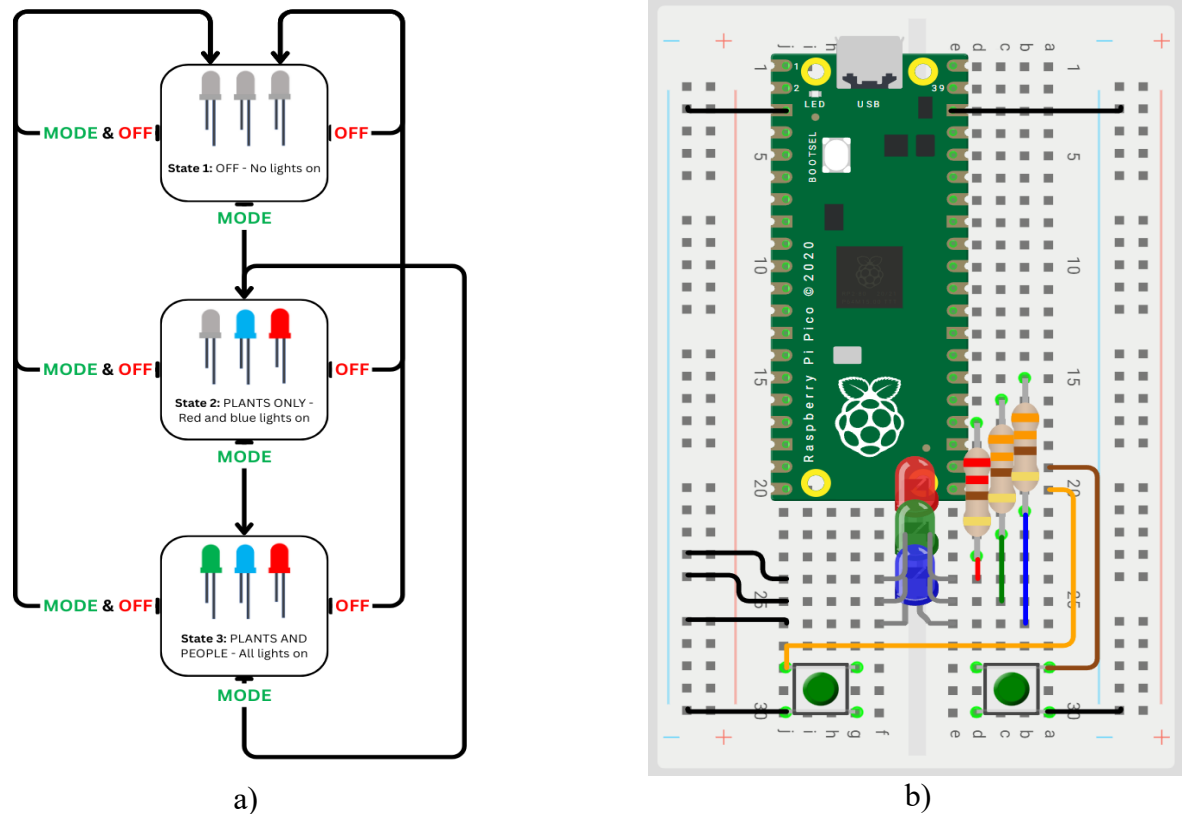


Fig. 2. a) State diagram of prototype logic that was translated to b) a physical circuit.

3. Results

The final prototype successfully demonstrated the required functionality (see 04.4), though with some trade-offs identified during verification.

- The pivot to the **bolted design** resulted in a **highly robust** body. While the original magnetic design would have offered faster disassembly, the nuts and bolts provide immense structural stability. The **custom buttons** provide pleasant tactile feedback.

4. Conclusion

For future design work, we suggest two improvements to address identified trade-offs:

1. A translucent diffuser layer above the LEDs to blend the RGB output, solving the color separation issue.
2. Replace the standard nuts with **threaded inserts** to combine the rigidity of the bolted design with the assembly speed of the magnetic concept.

2. Summary of Key Decisions

Software Subsystem

- A state diagram was made to outline the intended logic of the system. This allowed us to ensure that all inputs would be handled correctly in each state of the system (see 04.2_Software_StateDiagram).
- All code was written in Thonny IDE to transfer to the Raspberry Pi.
- The final code features descriptive comments and uses a distinct function to set the LED values to provide more clarity when reading the code (see 04.2_Software_SourceCode).

Electrical Subsystem

- Before cutting wires, a plan of the final circuit was sketched (see 04.2_Electrical_Schematic).

Structural Subsystem

- Fusion was used to develop the cad model of the outer shell.
- The dimensions and positions of the breadboard and buttons were measured to create a representation of these components in the software. This helped ensure that there would be no interference between the outer shell and these components when integrated into real life.
- Initially, it was intended to attach the upper and lower pieces of the outer shell using magnets. This material was difficult to obtain, so M3 bolts were used instead that passed vertically through the corners of the shell and fixed in place with a nut.
- The depth of the outer shell was designed to accommodate substantial clearance for components during testing.
- To ensure that the user could still press the buttons with ease, button caps were designed to rest on supports that protruded out of the front facing wall (see Fig. 1).

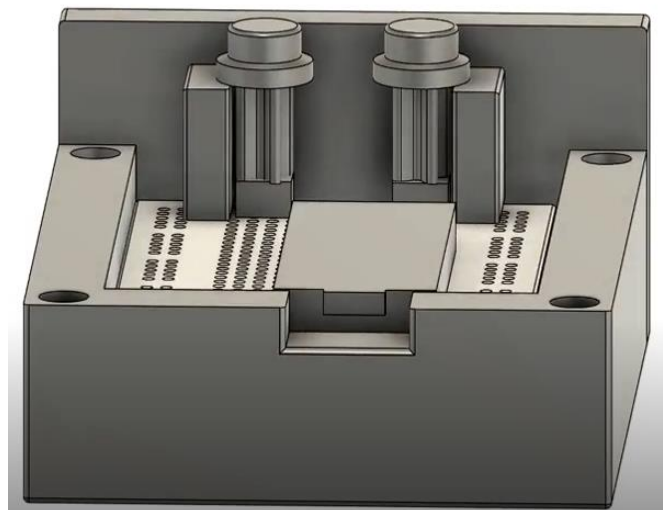


Fig. 1. Button caps that allow the user to press the buttons, despite the depth of the box.

- This wall also held the breadboard in place from below and above (see Fig. 2).

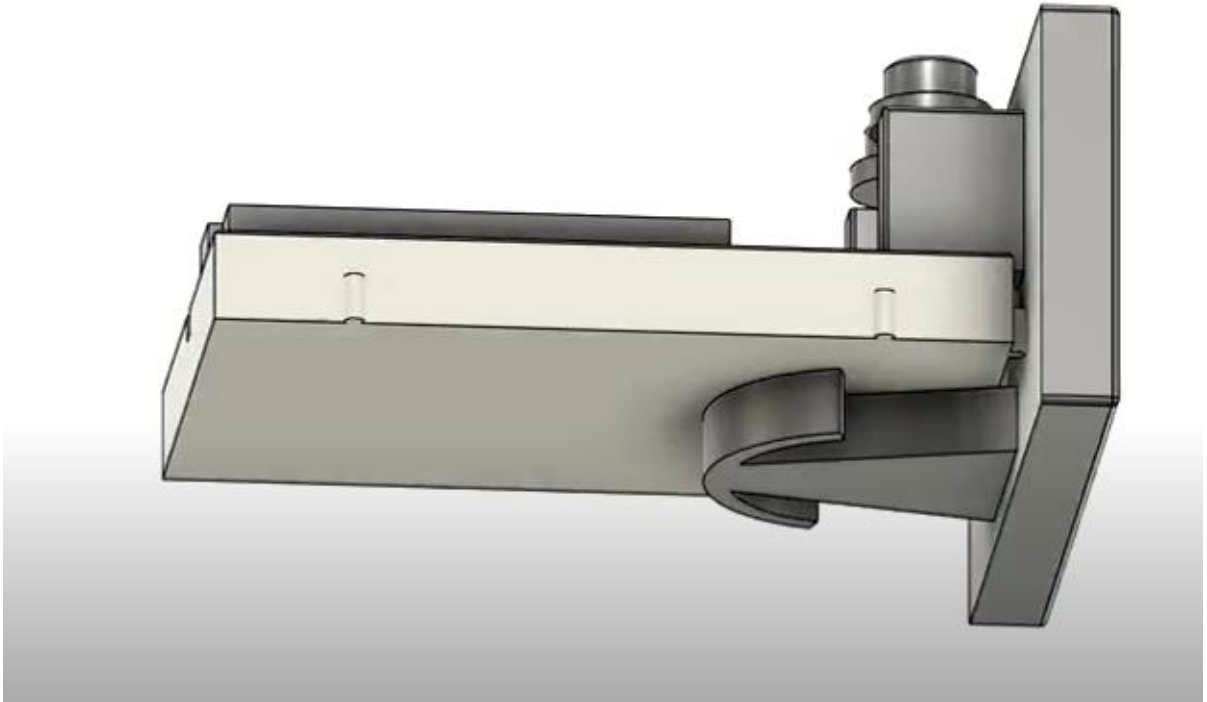


Fig. 2. The front facing wall “sandwiches” the breadboard to prevent it from moving up and down. The lower support is shaped to attach to the base of the outer shell.

- The disconnect between the upper and lower pieces of the model is only slightly higher than the surface of the breadboard to allow access to electrical components (the wall height relative to the breadboard surface is visible in Fig. 1).
- Characteristic lengths were increased by 0.1-0.2mm to account for the material shrinkage as it cools down from printing.
- A short animation was made in Fusion to communicate how each piece of the printed structure would fit together (see 04.2_Structural_CAD_Assembly).
- To reduce the piece count, this model was realized in PLA using a Prusa Mini. 3D printing was chosen to reduce the piece count of the design, as more complex shapes could be fabricated without the need to glue afterwards. With this method, only five parts were printed. In comparison, we anticipated that the number of parts would double if we used laser cutting (e.g. laser cutting the six walls would require

at least six components, while only three components were required using 3D printing). This simplifies the assembly and robustness of the final design.

3. Verification Protocol & Results

This spreadsheet is adapted from an example Verification and Validation Plan provided by NASA at https://www.nasa.gov/reference/appendix-d-requirements-verification-matrix/					
Specification ID	Statement of Related Specification for Prototype System	Verification Method (i.e., Test, Inspection, Analysis, Demonstration)	Verification Procedure (including metrics, if applicable)	Results	Notes
Electrical Subsystem					
O-1.1-ELEC-1	Shall provide distinguishable illumination when operating in an ambient-light environment, such that a visual distinction is made when illuminated elements are ON vs OFF in an ambient light (outdoor) environment between sunup and sunset.	Demonstration	1. Place the prototype in a well-lit indoor environment (approx. 500 lux). 2. Stand 1 meter away from the device. 3. Toggle the device power between ON and OFF. 4. Visually confirm that the difference between the illuminated state and the off state is clearly distinguishable.	All three states of the prototype are each distinguishable under these conditions. See "V1_Distinguishable_Illumination" under 04.4_Verification for demonstration.	
O-1.2-ELEC-1	Shall include at least 3 different LEDs of distinct colours (representing red, green and blue or another set of three distinct colours that are mappable to these).	Inspection	1. Count the number of LED components on the breadboard. 2. Inspect the color of each LED. 3. Confirm there are at least 3 distinct colors present (e.g., Red, Blue, Green).	Through the video "V2_Genera Usage" under 04.4_Verification, it is clear that there are 3 LEDs, each a distinct colors (Red, Blue, Green).	
O-3.1-ELEC-1	Shall use meaningful wire colours and a neat layout to ensure ease of debugging and repair.	Inspection	1. Inspect wiring layout. 2. Trace Power (3.3V) and Ground lines to verify consistent color usage (e.g., Red/Black, Dark/Light). 3. Check all signal wires to confirm that no two adjacent wires share the same color.	By inspecting the image "I1_Internals" under 04.4_Verification, it is confirmed that a neat wiring layout is used where all wires are visible. A consistent colour system is used as well: all wires leading to the ground bus on the right side of the breadboard are black; all wires connecting to pins on the microcontroller are coloured (e.g. red and green are used to connect the buttons, matching the colours used in the state diagram image "State_Diagram.png" under 04.2_build/evidence).	
Structural Subsystem					
O-1.2-STRU-1	Illuminated elements shall be visible to an observer when facing the device head-on.	Inspection	1. Place the device on a flat surface. 2. Position observer directly in front of the device. 3. Turn device on. 4. Confirm that light source is directly visible and unobstructed.	Through the video "V2_Genera Usage" under 04.4_Verification, it is shown that the LEDs are placed above the buttons, allowing for an unobstructed view while using the device head-on.	
O-3.1-STRU-1	Control inputs shall be accessible without disassembling or opening the structure.	Demonstration	1. With device fully assembled, attempt to press each button. 2. Confirm that buttons are actuated comfortably without tools or disassembly.	In the video "V2_Genera Usage" under 04.4_Verification, comfortable usage of the buttons without disassembling or opening the structure is demonstrated.	
O-3.1-STRU-2	The micro-USB cable shall be able to be easily connected and disconnected from the microcontroller board.	Demonstration	1. Hold the assembled and powered device steady with one hand. 2. Remove the micro-USB cable from the port. 3. Reinsert the cable into the port. 4. Measure the time; confirm each action takes < 2 seconds and requires no disassembly or excessive force.	The ease of access to the micro-USB port on the microcontroller board is demonstrated in the video "V3_Cable_Connection" under 04.4_Verification. It is clear by this demonstration that the outershell does not impede the usability of this port.	

O-3.1-STRU-3	The overall structure shall enable disassembly and re-assembly to gain access to the microcontroller and circuit elements without damage.	Demonstration	1. Begin with the fully assembled device. 2. Perform the disassembly procedure to take apart the device. 3. Confirm that circuit elements and the microcontroller are easily accessible. 4. Re-assemble the device. 5. Confirm that no parts were damaged (e.g., stripped screws/broken tabs) and that no permanent adhesives prevented the process.	The disassembly and re-assembly of the device is demonstrated in the video "V4_Disassembly&Assembly" under 04.4_Verification. It is shown through this demonstration that the user can perform these processes to gain full access to the circuit elements inside, without causing any damage to the prototype.	
O-4.1-STRU-1	All wiring and bare breadboard regions shall be covered.	Inspection	1. Fully assemble the device. 2. Verify that no bare wires or breadboard holes are visible through the casing gaps.	Through the video "V4_Disassembly&Assembly" under 04.4_Verification, all sides of the device are inspected, verifying that all wiring and bare breadboard regions are covered.	
O-4.2-STRU-1	The breadboard shall be securely contained such that the breadboard doesn't shift around when shaken or rotated.	Test	1. Hold the fully assembled device in hand. 2. Shake the device with moderate force (approx. 2 shakes/second) for 5 seconds. 3. Listen for the sound of loose components (rattling) or shifting mass inside the case.	This test is recorded in the video "V5_Shake_test" under 04.4_Verification. In all principal directions and axes of rotation, there is no clear sound that would indicate loose components within the device.	
Software Subsystem					
O-1.1-SOFT-1	There shall be a 'PLANTS ONLY' state where the red and blue LEDs are ON.	Demonstration	1. Power on the device. 2. Press the 'Mode' button to cycle to the 'Plants Only' state. 3. Visually inspect the LEDs. 4. Confirm that only the Red and Blue LEDs are illuminated.	In the video "V2_General_Usage" under 04.4_Verification, it is demonstrated that the "PLANTS ONLY" state functions as intended.	
O-1.2-SOFT-1	There shall be a 'PLANTS & PEOPLE' state where all three LEDs are ON.	Demonstration	1. From the 'Plants Only' state, press the 'Mode' button again. 2. Visually inspect the LEDs. 3. Confirm that the Red, Blue, and Green LEDs are all illuminated (creating a white-ish composite light).	In the video "V2_General_Usage" under 04.4_Verification, it is demonstrated that the "PLANTS & PEOPLE" state functions as intended.	
O-2.1-SOFT-1	There shall be an 'OFF' state where all three LEDs are OFF.	Demonstration	1. Press the 'Power' button while in the ON state to enter the OFF state. 2. Visually inspect the LEDs. 3. Confirm that zero LEDs are illuminated.	In the video "V2_General_Usage" under 04.4_Verification, it is demonstrated that the "OFF" state functions as intended.	
O-3.1-SOFT-1	The LED state shall not change unless a user presses a button.	Test	1. Set the device to 'Plants Only' mode. 2. Remove hands from all buttons. 3. Observe the device for 30 seconds. 4. Confirm the light state does not change autonomously. 5. Repeat steps 2-4 while device is set to 'Plants&People,' and then 'Off.'	The tests performed in videos "V6_Plants_30s", "V7_Plants&People_30s" and "V8_Off_30s" under 04.4_Verification provides sufficient verification that the device will not change states unless a button is pressed.	
O-3.1-SOFT-2	The LED state shall change when a button is pressed.	Test	1. Prepare to press the 'Mode' button. 2. Press the button and simultaneously watch the LEDs. 3. Confirm the light color changes immediately (latency < 0.5 seconds) after the physical press.	Through the video "V2_General_Usage" under 04.4_Verification, the responsiveness of the buttons is demonstrated as well as the a demonstration that all user inputs are handled as intended according to the state diagram image "State_Diagram.png" under 04.2_build/evidence.	
O-3.1-SOFT-3	There shall be no states achieved by the LEDs other than the 'OFF', 'PLANTS ONLY', and 'PLANTS&PEOPLE' states described in O-1.1-SOFT-1, O-1.2-SOFT-1, and O-2.1-SOFT-1 regardless of user input.	Test	1. Press each button rapidly 2. Verify that the device only ever displays the three defined states (OFF, Plants, Plants+People) and never settles on an undefined combination (e.g., Green only).	This is tested in the video "V9_Button_Reliability" under 04.4_Verification. Over numerous combinations of button presses, the device only ever displayed the three defined states.	